

Speech-Based Emotion, Age and Gender Detection with Speech-to-text

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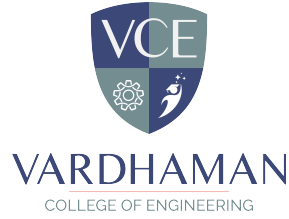
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CERTIFICATE

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Abstract

Since speech signals provide a wealth of information about the type of emotion, age, gender and content being spoken by the person speaking, automatic speech analysis is an important area of research to be applied to applications such as human computer interaction, intelligent voice system, and medical monitoring system. Still, identifying some of the features of the speaker on the basis of acoustics is not so easy due to the voice variations and the conditions under which the recording was done. The proposal in this paper is a hybrid machine learning/deep learning system, whereby, the emotions, the age, and the gender will be classified using speech with the aid of speech-to-text conversion. The system uses RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and song) during training and assessment. This is done through a multiplicity of acoustic features including MelFrequency Cepstral Coefficients (MFCC), delta and delta-delta coefficients, chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, spectral centroid, spectral rolloff and pitch features using the assistance of the Librosa library. A deep learning model that utilizes Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) networks to track both the spatial and the temporal speech signal features is used in the case of emotion recognition. In the instance of gender and age classification, the XGBoost (Extreme Gradient Boosting) classifiers are used in order to know well-structured features representations. There is also a speech-to-text application based on Google Speech Recognition API which converts the audio speech to text. The obtained outcomes of the experiment confirm that the hybrid framework, suggested by the authors, provides potentially effective results of emotions, gender, and age prediction, which makes it possible to conclude about the effectiveness of machine learning and deep learning techniques integration into the single speech analysis systems.

Keywords: Speech Emotion Recognition, Age prediction, Gender detection, machine learning, deep learning, CNN-BiLSTM, XGBoost, and Speech-to-Text

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Abbreviations

Abbreviation	Description
VCE	Vardhaman College of Engineering
AIML	Artificial Intelligence and Machine Learning

CHAPTER 1

Introduction

1.1 Introduction

Speech is one of the most natural ways of communication among human beings and there is a lot more information about speech compared to verbal communication. Voice of the individual can assist in determining his/her emotional state, age as well as sex among other key attributes of comprehending human behavior and how human-machines interactions can be enhanced. Automatic speech analysis is a research topic that is useful because voice technologies and artificial intelligence are developing quickly. The systems that could deduce the feeling and demographic facts such as virtual assistants, calls center analysis, medical surveillance and human-computer interaction are applicable with emotion and demographic speech recognition. Nevertheless, speech cues are hard to detect several traits of the speakers. The speech could differ greatly as compared to the accent of the person speaking, the way he is talking, the conditions in which speech is recorded and the noises around him [1]. The reality that such diversities means that the conventional systems will fail to decode speech data with accuracy of any kind. The speech analysis systems can thus be enhanced by inculcating the machine learning and deep learning technologies. The paper creates a system that assumes the automated analysis of speech and the recognition of various traits of a speaker, using an audio signal. The system is oriented to the perception of the feeling, gender, age group as well as the translation of the speech into the text. In order to accomplish this goal, the given system has a huge number of critical parts. Speech-Based Emotion, Age, and Gender Detection with Speech-to-Text is an advanced interdisciplinary system that combines techniques from Speech Processing, Machine Learning, and Natural Language Processing to extract meaningful information from human speech. The primary objective of this

system is to analyze audio signals not only to understand what is being said through speech-to-text conversion, but also to interpret who is speaking and how they are feeling. Speech signals carry rich paralinguistic information such as tone, pitch, intensity, and speaking rate, which can be used to detect emotions like happiness, sadness, anger, or neutrality. Simultaneously, acoustic features such as fundamental frequency, formants, and spectral characteristics help in estimating the speaker's age group and identifying gender.

The integration of speech-to-text plays a crucial role by converting spoken language into textual form, enabling further semantic and contextual analysis. This allows the system to combine both audio-based features and text-based insights for improved accuracy. Modern approaches rely on deep learning models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks to automatically learn complex patterns from speech data. These models are trained on large annotated datasets to recognize variations in voice and language. Such systems have wide applications in human-computer interaction, virtual assistants, call center analytics, healthcare monitoring, and security systems. By enabling machines to understand not only speech content but also emotional and demographic cues, this technology significantly enhances the naturalness and effectiveness of communication between humans and intelligent systems.

- The project Speech-Based Emotion, Age, and Gender Detection with Speech-to-Text is meant to create a smart system that can analyze speech of a person so as to identify the emotion of that person, the age bracket, and gender of the individual as well as converting the speech to a text format.
- The system accepts speech input and obtains significant audio data including pitch, tones, frequency and speech energy, which are used in comprehending the nature of the speaker.
- Emotions (happy, sad, angry, or neutral) and the age group and the gender of the speaker are classified with the help of the Machine Learning or Deep Learning models. The importance of this research is that the speech has

not only content information but also emotional cues, which are capable of enhancing human communication with computers and enhancing the human-computer interaction

- The technology is applicable in domains like call center analysis, health-care tracking, virtual assistants, security systems and customer sentiment analysis. The overall objective of the project is to create an effective system that would be able to analyze speech signals automatically and give precise results in the identification of emotion, age, and gender as well as speech transcription. The project scope will cover processing of speech data, feature extraction, model training, speech-to-text integration, presentation of the findings on a barebone interface or a web dashboard

1.2 Background and Motivation

The high development of online resource of recruitment has simplified job search among millions of individuals on the global front. It has, however, also led to several fraudulent job advertisements made by scammers to gather personal information or money by job seekers. Automated systems that can detect such fraudulent job posts are being formulated in the sphere of Artificial Intelligence, Natural Language Processing (NLP), and Cybersecurity. Previous literature has employed machine learning algorithms like Logistic Regression, Support Vector Machines or any of the random forest with text-processing methods like TF-IDF and Bag-of-Words to classify job posts in real or fake. The recent developments in the field of deep learning and explainable AI, including SHAP and LIME, have been proposed to enhance the accuracy and transparency of prediction. In spite of all these developments, most of the systems that are present do not have real-time detection, multi-lingualism, and easily accessible interfaces. Thus, the offered project is a proposal of AI-based fake job posts detection and verification system with a explainable NLP and a web interface that can be used by users to simply check what a job posting is considered a fake or a real one and why a specific post is defined as fake. The project is deemed to meet the increasing demand of quality job-fraud

detecting tools and helps in enhancing the security of online recruitment.

The rapid growth of online recruitment platforms has made job searching easier worldwide, but it has also increased the number of fraudulent job advertisements. Scammers often create realistic-looking job posts to collect personal information or money from job seekers. To address this issue, Artificial Intelligence, Natural Language Processing (NLP), and Cybersecurity techniques are used to develop automated detection systems.

Earlier methods used machine learning algorithms like Logistic Regression, Support Vector Machines, and Random Forest with techniques such as TF-IDF and Bag-of-Words. Recent advancements in deep learning and Explainable AI methods like SHAP and LIME have improved accuracy and transparency by providing reasons behind predictions. However, many existing systems still lack real-time detection, multilingual support, and user-friendly interfaces.

Additionally, fraudsters continuously evolve their techniques, making detection more challenging. Therefore, the proposed system focuses on developing an AI-based fake job post detection model with explainable NLP and a simple web interface. This system aims to provide accurate, real-time, and transparent results, helping users identify fake job postings and improving the security of online recruitment platforms.

- Online job portals have increased opportunities but also job fraud.
- Jobs advertised in a fraudulent manner are used to defraud the job seekers with money and personal information.
- Several systems are not fitted with real time detection and easy to use dashboards.
- his project creates the AI-based verification system that has explainable outcomes.

1.3 Objectives of the Project Work

The main objective of this project is to develop an intelligent AI-based system capable of accurately identifying whether a job posting is real or fake

using advanced machine learning and Natural Language Processing (NLP) techniques. The system aims to analyze job descriptions, company details, and recruiter information to extract meaningful patterns and improve prediction accuracy through modern models. It also focuses on incorporating Explainable AI methods such as SHAP or LIME to provide clear and understandable reasons behind each prediction, thereby increasing user trust and transparency.

In addition, the project emphasizes real-time detection so that users can instantly verify job postings before applying. It also aims to support multilingual inputs, making the system accessible to a wider range of users. A simple and user-friendly web interface will be developed to ensure ease of use, even for non-technical users. Furthermore, the system is designed to be scalable and continuously updated with new data so that it can adapt to evolving fraud patterns. Overall, the project seeks to reduce online job scams, protect users from financial and data loss, and contribute to building a safer and more reliable digital recruitment environment.

The primary objective of the project is to create a smart system that can automatically identify and test fraudulent job postings with the help of the Artificial Intelligence and Natural Language Processing technologies. The aim of the project is to enhance the credibility of internet recruiting network and safeguard the job applicants against frauds. The objectives of this work are the following ones:

- To create an AI-based model, which will be able to categorize job posts as either real or fake using Natural Language Processing techniques.
- To evaluate various machine learning and deep learning algorithms to determine the most predictive one in determining a fraudulent job advert.
- To create a explainable AI component, which will give a clear explanation as to why a job advertisement is considered fake or genuine.
- To develop a user friendly web dashboard where people can post or identify job postings as legitimate.

1.4 Organization of the Report

The final section of Chapter 1 provides an outline of the structure of the entire report. It explains the organization and flow of the chapters and gives the reader an understanding of how the content will be presented. This section can be a simple list or paragraph summarizing each chapter's main points.

- Provide a brief description of what each chapter will cover.
- Typically, the structure would include:
 - **Chapter 2:** Literature Review
 - **Chapter 3:** Technical Title Related to the Prior Research
 - **Chapter 4:** Technical Title Related to the Proposed Work
 - **Chapter 5:** Results and Discussion
 - **Chapter 6:** Conclusions and Recommendations

CHAPTER 2

Literature Survey

2.1 Introduction

One of the most natural humans means of communication is speech. It does not only transfer words but has the significance of information just like emotion, age, and gender of the speaker. As technologies of artificial intelligence and machine literacy have evolved, experimenters have focused on the creation of systems that have the capability to automatically dissect speech signals in order to reward similar information. Grounded emotion, age, and gender discovery models have emerged as a key discovery area in related directions as mortal- computer commerce, virtual sidekicks, security systems and medical surveillance.

several papers have been done on various methods of testing speech cues and transforming them to informative data. The speech to textbook technology in general is applied to convert spoken language into written textbook which also can be anatomized with the help of natural language processing ways. Meanwhile, the aural features similar in pitch, tone and frequency are employed to recognize the emotional countries and determine the demographic features such as age and gender. Colorful machine literacy and deep literacy algorithms (with support vector machines, convolutional neural networks, and intermittent neural networks) have been used by experimenters to reduce the delicacy of these discovery systems.

This literature review is meant to break down prior exploration on speech-grounded emotion, age, and gender discovery and to learn the various styles that are employed by the experimenters. The study of being work allows defining the advantages and shortcomings of the contemporary solutions and finding out how they can be improved. This review lays a ground upon the formation of a more ideal framework integrating speech- to- textbook

technology with machine literacy methods of direct articulation of emotion, age, and gender, based on the speech cues.

The literature survey plays a crucial role in understanding the existing research and developments related to fake job post detection. With the rapid growth of online recruitment platforms, the problem of fraudulent job postings has gained significant attention among researchers. Various studies have been conducted using techniques from Machine Learning, Natural Language Processing (NLP), and data mining to identify and classify fake job advertisements. This chapter provides a comprehensive review of these approaches and highlights their contributions and limitations.

Earlier research primarily focused on traditional machine learning algorithms such as Logistic Regression, Support Vector Machines (SVM), and Random Forest for classification tasks. These methods utilized text-based features extracted using techniques like Bag-of-Words and TF-IDF to analyze job descriptions. While these approaches achieved reasonable accuracy, they often failed to capture deeper contextual meaning and semantic relationships within the text.

With advancements in technology, recent studies have shifted towards deep learning models, including Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and transformer-based models. These models are capable of learning complex patterns from large datasets and provide improved performance in detecting fraudulent content. Additionally, researchers have started integrating Explainable AI techniques such as SHAP and LIME to make model predictions more transparent and interpretable.

Despite these improvements, several challenges remain. Many existing systems lack real-time detection capabilities and are limited to specific datasets or languages. Moreover, the absence of user-friendly interfaces makes these systems less accessible to common users. Therefore, this literature survey helps in identifying research gaps and motivates the development of a more efficient, accurate, and user-oriented fake job post detection system.

2.2 Review of Prior Research

Various experimenters have performed speech analysis experiments to infer feelings, age, and gender by the voice cues. Such studies are meant to alleviate mortal- computer commerce by allowing the computers to comprehend the mortal speech features more efficiently.

The initial research in speech emotion recognition gave significant attention to basing aural characteristics that are like in pitch, energy, and Mel- frequency Cepstral Portions(MFCC). These characters assist in comparing emotional nations such as joyfulness, sadness, wrathfulness and being unbiased. These features were generally used in classifying feelings using machine literacy algorithms such as the Support Vector Machines(SVM), k- Nearest Neighbors(KNN), and Decision Trees.

later research papers proposed the use of deep literacy in the vein of Convolutional Neural Network(CNN) and intermittent Neural Network(RNN) to alleviate the construal nature of speech analysis systems. Such models can learn complicated patterns on their own by speech data and they are more effective than standard machine literacy styles.

Gender and age discovery using speech signals has also been experimented by experimenters. The process of gender discovery is usually done through testing of the variation in voice in terms of pitch, frequency range and mouth attributes of a manly and womanish speaker. On the other hand, age discovery is the process of measuring the speech patterns and other features of the mouth which transform with age of an individual. colorful machine literacy models have been used to estimate the age groups using recordings of the speech.

The other major field of research is the speech- to- textbook technology which translates the spoken words or language into written textbook. The technology finds widespread application in voice sidekicks, recap systems and availability tools. Experimenters have been able to design integrated systems using speech- to- textbook systems and speech point analysis to be able to not only decode spoken material, but also determine similarities between speakers such as emotion, age, and gender.

Despite the promising outcomes of the many studies done on exploration, the issues of high delicacy in real- world conditions such as background noise, other accentuations and speaking styles variations still pose a challenge, therefore, further exploration is required to produce stronger and more accurate models to use in discovery systems that are based on speech, age and gender.

2.3 Identified Research Gaps

Even though many experimenters have performed their research on speech analysis and its ability to detect emotion, age, and gender, the current systems have a number of limitations. Through a literature review, some of the gaps that need to be explored have been associated.

A significant of the major gaps is the scarcity of delicacy in real-world environment, many models are trained on controlled datasets with clean audio recordings, however, in practice, speech signals can include background noise, varying accentuations, and speaking style variations, which decrease the delicacy of discovery systems.

The other gap in exploration is the individual examination of emotion, age, and gender in many studies. utmost exploration is only to identify one or two features of speech instead of building a single system that can describe all three features simultaneously. This is because it is difficult to integrate emotion, age and gender discovery in a single system.

also, there are many systems computing considerably on aural features of speech, but smaller studies have used aural features with speech- to- textbook analysis. Combining the content of speech with aural parcels may alleviate the functioning of discovery systems, yet no significant research has been done in this field.

The other shortcoming that is realized in previous exploration is the absence of various and extensive datasets. many models are developed on small datasets, which might not take into account different languages, accentuations, as well as age groups. This influences the conception ability of the models when subjected into the real world druggies.

So there is a necessity to come up with a stronger mechanism that will

combine speech- to- textbook technology with machine literacy methods of more directly describing emotion, age, and gender. The suggested design will fill these exploration gaps with the objective of creating a useful speech-grounded discovery system that can be successful in real scripts.

2.4 Summary

The literature check discussed in this chapter has reviewed colorful exploration workshop related speech- based emotion, age, and gender discovery systems. past literature has demonstrated that speech cues carry significant aural coefficients which can be compared as pitch, tone, frequency, and Mel-frequency Cepstral Portions(MFCC), and can be used to recognize various emotional nations and speaker attributes.

Many experimenters have used equivalent machine literacy methods such as Support Vector Machines, k- Nearest Neighbors and Decision Trees when performing speech bracket tasks. More recently, some deep literacy techniques such as Convolutional Neural Networks and intermittent Neural Networks have also seen widespread application in order to alleviate the frailty of speech analysis systems. These sophisticated models can learn sophisticated patterns using speech data and they have demonstrated improved performance over conventional styles.

The check also emphasized the role of speech- to- textbook technology in transforming the spoken language into the textual form which can also be anatomized through ways of natural language processing. A synthesis of a speech recognition and aural point analysis assist in building more intelligent systems, which are capable of interpreting the content and the nature of the speech.

Nonetheless, there were a few disadvantages associated with it as exploration, such as the problem of noisy environment, inadequate datasets, lack of built-in systems that could detect emotion, age, and gender simultaneously. These loopholes demonstrate the necessity of more powerful and successful models.

In general, the literature review presents a solid background of the proposed

system and assists in choosing appropriate options on how a system based on speech can be developed, integrating the speech- grounded emotion, age and gender discovery system with speech- to- textbook technology.

CHAPTER 3

Title Related to Prior Research

3.1 Introduction

Speech is one of the most natural and the strongest types of human communication which can not only consist of linguistic data but also of paralinguistic elements like emotion, age, and gender. Over the last several years, the progress of artificial intelligence and machine learning allowed creating systems that are capable of automatically analyzing speech signals to identify these properties. Speech-based emotion, age and gender recognition deals with audio input processing to determine the emotions of the speaker, a rough estimate of their age bracket and gender based on aspects of acoustic audio such as sound tone, pitch, frequency and speech patterns. These systems can turn spoken language into written text and at the same time analyze the inherent speaker characteristics when combined with the speech-to-text technology. The combination is better applicable in human-computer interaction, virtual assistant, medical monitoring, security, and customized user experiences. Previous study on the same field has been aimed at enhancing the feature extraction methods, classification algorithms alongside multimodal methods in order to gain more accurate and stronger results in real world conditions.

Various approaches have been explored in earlier works, primarily using machine learning algorithms such as Logistic Regression, Support Vector Machines (SVM), and Random Forest. These models rely on structured and unstructured data extracted from job postings, including job descriptions, company profiles, and required skills. Text preprocessing techniques such as tokenization, stop-word removal, stemming, and feature extraction methods like TF-IDF and Bag-of-Words play a vital role in converting textual data into a suitable format for model training.

In addition to traditional machine learning methods, recent research has emphasized the use of deep learning models for improved performance. Techniques such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are capable of capturing sequential patterns and contextual relationships in text data. Furthermore, transformer-based models have shown significant improvements in handling large-scale textual information.

Another important aspect of prior research is the incorporation of Explainable AI techniques, such as SHAP and LIME, which provide insights into model predictions. These methods help users understand the reasoning behind classification results, thereby increasing trust and transparency in the system.

3.2 Subtitle of Existing Work

The current research on speech-based emotion, age, and gender detection have received a lot of interest owing to the development of machine learning and speech processing algorithms. Previous studies mainly concentrated on the extraction of acoustic properties including pitch, energy, Mel-Frequency Cepstral Coefficients (MFCCs) and spectral properties to determine speaker properties. Conventional approaches involved classifiers such as the Support Vector Machines (SVM) and the Hidden Markov Models (HMM). Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks are deep learning techniques that have gained popularity over the last few years to achieve accuracy and handle speech patterns of considerable complexity. Also, these systems have been equipped with speech-to-text technology in order to transform spoken language into text so as to achieve a better understanding of context and analysis of the multimodal context. In spite of these technology, issues of background noise, language dependency and differences in speech quality continue to impact the performance of the system. On the whole, the current research has provided a solid base, and it is possible to advance to making more powerful and real-time systems.

Existing work in the area of fake job post detection has primarily focused on applying machine learning and Natural Language Processing (NLP) techniques

to classify job advertisements as genuine or fraudulent. Early studies utilized traditional classification algorithms such as Logistic Regression, Support Vector Machines (SVM), and Random Forest, which relied on manually engineered features extracted from job descriptions. Techniques like Bag-of-Words and TF-IDF were commonly used to convert textual data into numerical form for model training. These approaches provided a baseline for detection but were limited in capturing the deeper context and semantics of language.

In recent years, researchers have explored advanced methods using deep learning models such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and transformer-based architectures. These models are capable of understanding sequential and contextual relationships in text, leading to improved accuracy in detecting fraudulent job postings. Additionally, some studies have incorporated metadata features such as company information, salary details, and posting patterns to enhance model performance.

Another important advancement in existing work is the use of Explainable AI techniques like SHAP and LIME, which help interpret model predictions by identifying the most influential features. This has improved the transparency and reliability of detection systems. However, many existing solutions still face challenges such as lack of real-time implementation, limited multilingual support, and absence of user-friendly interfaces. These limitations highlight the need for more robust and accessible systems, which motivates the development of the proposed approach.

3.3 Speech Feature Extraction

The extraction of speech features is an important procedure in the system proposed. It entails the process of transforming raw speech signals to meaningful numerical features, which are subject to analysis. Identical features that are commonly used are Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy and spectral features. Such characteristics can represent key attributes of speech including frequency, tone and intensity which are significant in the proper identification of emotion, age and gender.

Speech feature extraction is a fundamental step in speech-based systems,

where important information is derived from raw audio signals for further analysis. In the context of emotion, age, and gender detection, speech signals contain valuable acoustic and prosodic features such as pitch, tone, energy, and speaking rate. These features help in understanding not only the content of speech but also the characteristics of the speaker.

The process begins with preprocessing of the audio signal, which includes noise removal, normalization, and segmentation of speech into smaller frames. After preprocessing, various features are extracted from each frame. Commonly used features include Mel-Frequency Cepstral Coefficients (MFCC), which represent the short-term power spectrum of sound and are widely used in speech recognition tasks. Other important features include pitch (fundamental frequency), which is useful for gender detection, and energy or intensity, which can indicate emotional states such as anger or excitement.

In addition, spectral features such as spectrograms and chroma features provide a visual and mathematical representation of frequency variations over time. Temporal features like zero-crossing rate and speaking rate also contribute to identifying speech patterns. These extracted features are then used as input to machine learning or deep learning models for classification and prediction tasks.

Overall, effective speech feature extraction improves the performance of the system by capturing both linguistic and paralinguistic information from speech, making it possible to accurately detect emotions, estimate age, and identify gender.

3.4 Classification Techniques

After features have been obtained out of the speech signal, the classification method is applied, and the required outputs are obtained. The machine learning algorithms that may be commonly used are a Support Vector Machine (SVM) or the deep learning models such as the Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM) network. Such models learn patterns according to the training data, and they classify the given speech in different groups such as emotions, ages and gender with great precision.

Classification techniques play a vital role in speech-based systems for identifying emotions, age, and gender from extracted speech features. After feature extraction, the processed data is fed into classification models that learn patterns and relationships to categorize the input into predefined classes such as emotional states (happy, sad, angry), age groups, or gender.

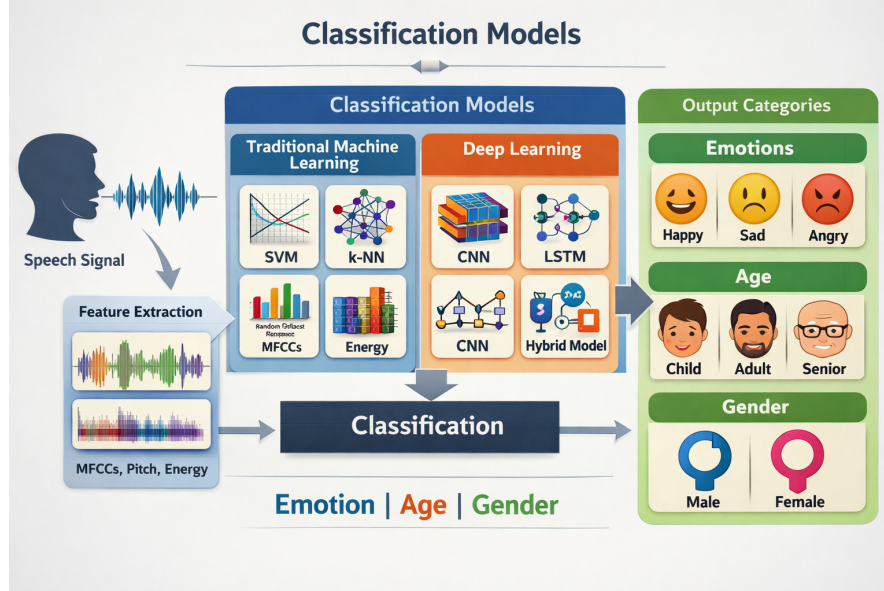


Figure 3.1: Classification Models

Traditional machine learning techniques such as Logistic Regression, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forest have been widely used for classification tasks. These models are effective when combined with well-engineered features like MFCC, pitch, and energy. They are relatively simple, computationally efficient, and provide good baseline performance. However, their ability to capture complex patterns in large and unstructured data is limited.

To overcome these limitations, deep learning techniques have been introduced in recent research. Models such as Convolutional Neural Networks (CNNs) are used to analyze spectrograms and extract spatial features, while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are effective in capturing temporal dependencies in speech signals. These models automatically learn high-level representations from raw or minimally processed data, leading to improved accuracy.

Furthermore, hybrid approaches that combine multiple models or integrate

both audio and text features (from speech-to-text) have shown better performance. Some systems also incorporate ensemble learning techniques to enhance robustness and reliability.

Overall, the choice of classification technique depends on factors such as dataset size, computational resources, and required accuracy. Advanced deep learning models generally provide better results, but traditional methods remain useful for simpler and faster implementations.

3.5 Speech-to-Text Integration

The speech-to-text module translates verbal words into written words. The module will improve the system capabilities as it allows text generation as well as speech analysis. It is specifically important to use it in voice assistants, transcription services, and accessibility applications. The speech-to-text and the classification model together will give a more detailed insight into the user input.

Speech-to-Text integration is an important component in speech-based systems, enabling the conversion of spoken language into written text for further analysis. In the context of emotion, age, and gender detection, this process helps in understanding not only the acoustic properties of speech but also the linguistic content conveyed by the speaker. By combining both audio features and textual information, the system can achieve more accurate and meaningful predictions.

The process begins with capturing the audio input and converting it into a digital signal. This signal is then processed using Automatic Speech Recognition (ASR) techniques to generate the corresponding text. Modern speech-to-text systems use deep learning models such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and transformer-based architectures to improve recognition accuracy, even in noisy environments.

Once the speech is converted into text, Natural Language Processing (NLP) techniques are applied to extract useful information such as keywords, sentiment, and context. This textual data can enhance emotion detection by identifying emotionally relevant words and phrases. It also supports better

interpretation of the speaker's intent and communication style.

Integrating speech-to-text with audio-based analysis creates a multimodal system that leverages both linguistic and paralinguistic features. This combination improves the overall performance and robustness of the system. Additionally, it enables applications such as voice assistants, customer support analysis, and real-time monitoring systems.

Overall, speech-to-text integration plays a crucial role in enriching the system's capabilities by bridging the gap between spoken and textual information, leading to more accurate and comprehensive analysis.

3.6 System Architecture

There are different steps involved in the system architecture and they include audio input, preprocessing, feature extraction, classification and the generation of output. Preprocessing of the speech signal involves the first step of capturing and removing noise. Subsequently, areas of interest are extracted and sent to emotion, age and gender classification models. Finally, speech-to-text option is utilized in order to transcribe the audio to text and display the results to the user.

The system architecture defines the overall structure and workflow of the speech-based emotion, age, and gender detection system integrated with speech-to-text functionality. It illustrates how different components interact to process input data and generate meaningful outputs. The architecture is designed in a modular manner to ensure scalability, flexibility, and efficient processing.

The system begins with the input module, where speech is captured through a microphone or uploaded as an audio file. This raw audio signal is then passed to the preprocessing module, which performs noise reduction, normalization, and segmentation to improve the quality of the input data. Cleaned audio signals are then forwarded to the feature extraction module, where important features such as MFCC, pitch, energy, and spectral characteristics are extracted.

These extracted features are then provided to the classification module, which uses machine learning or deep learning models such as CNN, RNN,

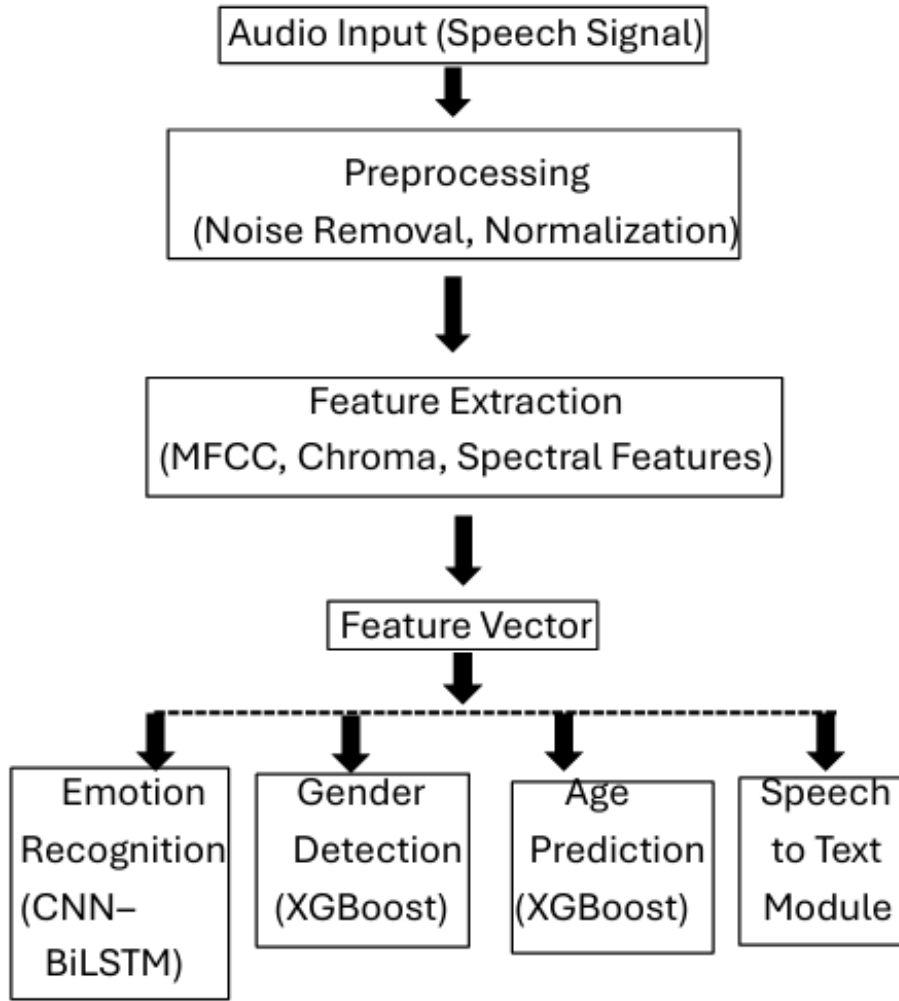


Figure 3.2: Proposed flow diagram

or LSTM to predict emotion, age group, and gender. In parallel, the audio signal is also processed by the speech-to-text module, which converts speech into textual form using Automatic Speech Recognition (ASR) techniques.

The generated text is further analyzed using Natural Language Processing (NLP) techniques to extract contextual and semantic information. Both audio-based predictions and text-based insights are combined in a fusion module to improve the accuracy and reliability of the final output.

Finally, the results are displayed through the output interface, which presents detected emotion, estimated age group, gender, and transcribed text in a user-friendly manner. This architecture ensures efficient data flow and enables real-time processing, making the system suitable for practical applications such as virtual assistants, customer feedback analysis, and human-computer

interaction systems.

3.7 Summary

The audio input, preprocessing and feature extraction and classification and output generation are some of the processes involved in the system architecture. The speech signal is first sampled and binned to take off the noise. Then, the features of interest are obtained and forwarded to classification models to recognize emotion, age and gender. Lastly, the speech-to-text application is supposed to convert the audio to the text and present the findings to the user.

This chapter presented an overview of the key concepts and techniques related to prior research in speech-based emotion, age, and gender detection systems. It discussed the importance of speech feature extraction methods such as MFCC, pitch, and energy, which help in capturing essential acoustic and prosodic characteristics of speech signals. Various classification techniques, including traditional machine learning models and advanced deep learning approaches, were also explored for accurate prediction of emotional states, age groups, and gender.

The chapter further explained the role of speech-to-text integration in converting spoken language into textual data, enabling additional analysis using Natural Language Processing (NLP) techniques. In addition, the system architecture was described to provide a clear understanding of how different components interact to process speech data and generate meaningful outputs.

Overall, this chapter highlights the strengths and limitations of existing approaches and establishes a strong foundation for the proposed system. It emphasizes the need for more accurate, real-time, and integrated solutions that combine both audio and textual features for improved performance.

This chapter provided a comprehensive overview of the prior research related to speech-based emotion, age, and gender detection systems. It discussed the key concepts such as speech feature extraction, classification techniques, speech-to-text integration, and system architecture that form the foundation of existing approaches.

The chapter highlighted the use of traditional machine learning methods as well as advanced deep learning models for analyzing speech signals and improving prediction accuracy. It also emphasized the importance of combining acoustic features with textual information obtained through speech-to-text conversion to enhance system performance.

Furthermore, the strengths and limitations of existing systems were analyzed, including challenges such as handling noisy data, limited accuracy in age prediction, and lack of real-time implementation in some cases. These observations helped in identifying research gaps and provided motivation for developing the proposed system.

Overall, this chapter established a strong technical background and justified the need for an improved, efficient, and integrated approach for speech-based analysis, which is addressed in the proposed work.

CHAPTER 4

Speech-Based Emotion, Age and Gender Detection with Speech-to-Text

4.1 Introduction

In this chapter, the suggested system that will automatically analyze the speech signals and extract significant information in the form of emotion, age, and gender of the speaker, as well as converting speech into text, are described. Speech contains non-linguistic and linguistic data, the analysis of these features assists in the interpretation of human behavior and pattern of communication. Incrementation of the integration of speech-to-text with classification tasks increases the usability of the system in real-world use like virtual assistants, healthcare monitoring, customer support systems and human-computer interaction. It is proposed that the proposed system can be used to offer an efficient, exact, and automated solution to the problem based on the combination of speech processing methods and machine learning and deep learning models. The design, working and implementation of the system are explained in this chapter.

Speech is one of the most natural and effective forms of human communication, carrying not only linguistic information but also emotional and personal characteristics of the speaker. In recent years, there has been growing interest in developing intelligent systems that can analyze speech signals to detect emotions, estimate age, and identify gender. These capabilities play an important role in enhancing human-computer interaction, making systems more responsive, personalized, and user-friendly.

Speech-based emotion detection focuses on identifying the emotional state of a speaker by analyzing features such as tone, pitch, energy, and speaking rate. Similarly, age and gender detection rely on acoustic and physiological characteristics present in the speech signal, such as fundamental frequency and

vocal tract features. These tasks require advanced processing techniques to accurately interpret both the explicit and implicit information embedded in speech.

The integration of speech-to-text technology further enhances the system by converting spoken language into textual form. This allows the use of Natural Language Processing (NLP) techniques to analyze the semantic and contextual meaning of the speech. By combining both audio-based and text-based features, the system can achieve better accuracy and more meaningful insights.

Recent advancements in machine learning and deep learning have significantly improved the performance of such systems. Models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks are widely used to capture complex patterns in speech data. These technologies enable the development of robust systems capable of real-time analysis and practical applications.

Overall, this chapter introduces the proposed system for speech-based emotion, age, and gender detection integrated with speech-to-text. It provides the foundation for understanding the methodology, design, and implementation of the system presented in the following sections.

4.2 Proposed System Architecture

The suggested system adheres to a systematic flow comprising of a series of steps. The audio input is the first step where the speech signal is captured using a microphone or uploaded in the form of an audio file. The second is preprocessing in which the noise reduction, normalization, and silence removal methods are used to enhance the quality of the speech signal. The system then undergoes preprocessing followed by feature extraction, when the important features of the speech signal are extracted including MFCC (Mel-frequency Cepstral Coefficients), pitch, energy, and spectral features.

These features are extracted and given to the classification step where machine learning or deep learning models are used to predict emotion, age group and gender based on the features. Models like CNN and LSTM are

mostly applied because they can capture spatial and temporal patterns of speech information. Concurrently the speech to text module changes the verbal data into textual information through speech recognition methods. Lastly, the system displays the perceived emotion, age range, the gender, and the respective text. Flexibility and scalability of the system is guaranteed by this modular architecture.

The proposed system architecture is designed to efficiently analyze speech signals for detecting emotion, age, and gender, along with speech-to-text conversion. The architecture follows a modular approach, where each component performs a specific function to ensure accuracy, scalability, and real-time performance.

The process begins with the input module, where speech is collected either through a microphone or as a pre-recorded audio file. This audio is then passed to the preprocessing module, which performs noise reduction, normalization, and segmentation to improve the quality of the signal. Clean and processed audio ensures better feature extraction and model performance.

Next, the feature extraction module extracts important acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and spectral features. These features capture both the physical and emotional characteristics of the speech signal. The extracted features are then fed into the classification module, where machine learning or deep learning models such as CNN, RNN, or LSTM are used to predict the speaker's emotion, age group, and gender.

In parallel, the audio signal is also processed by the speech-to-text module, which converts spoken language into text using Automatic Speech Recognition (ASR). The generated text is further analyzed using Natural Language Processing (NLP) techniques to extract contextual and semantic information.

The outputs from both audio-based classification and text-based analysis are combined in a fusion module to improve overall accuracy and reliability. Finally, the results are displayed through a user interface, which presents the detected emotion, predicted age group, gender, and transcribed text in an easy-to-understand format.

This architecture ensures an integrated and efficient system that leverages

both acoustic and linguistic features, enabling accurate and real-time speech analysis for various practical applications.

4.3 Methodology

The system proposed has a methodology which consists of a series of steps that have to be followed one by one. In the first place, the speech data are received on the basis of datasets or on-the-fly sources of input. Audio collected is disaggregated into small frames to guarantee that it becomes more efficient in analysis. The frames undergo processing in order to derive the significant features which are employed in capturing the properties of the speech signal. The importance of feature extraction is that it has a direct implication on the performance of the classification models.

After the extraction of the features, the data is divided into the training and the test set. The machine learning or deep learning models are then trained on the training data and they subsequently assist them in learning trends in regard to different feelings, age groups and gender. Evaluation of the models follows the training of the models by applying the test data to identify the effectiveness of the models. This procedure is complemented by the speech-to-text that interprets audio messages into text by using the automatic speech recognition technology. The methodology ensures that all the parts work in a co-ordinated way to provide the appropriate and legitimate results.

The methodology of the proposed system involves a sequence of well-defined steps to process speech data and accurately detect emotion, age, and gender along with speech-to-text conversion. The process begins with data collection, where speech samples are gathered either from publicly available datasets or real-time audio input through a microphone. These datasets contain labeled information for emotions, age groups, and gender, which is essential for training the models.

The next step is data preprocessing, where the raw audio signals are cleaned and prepared for analysis. This includes noise removal, normalization, and segmentation of audio into smaller frames to ensure consistency and improve the quality of input data. After preprocessing, feature extraction is performed

to obtain relevant acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and spectral features that represent the characteristics of speech.

Following this, the extracted features are used to train and test classification models. Machine learning and deep learning algorithms such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks are employed to classify emotion, age, and gender. The dataset is typically divided into training and testing sets to evaluate model performance and avoid overfitting.

In parallel, the system performs speech-to-text conversion using Automatic Speech Recognition (ASR) techniques. The converted text is further processed using Natural Language Processing (NLP) methods to analyze semantic and contextual information. This helps in enhancing emotion detection by considering both spoken content and vocal features.

Finally, the outputs from audio-based classification and text-based analysis are combined to produce the final result. The system then displays the detected emotion, predicted age group, gender, and transcribed text through a user-friendly interface. This step-by-step methodology ensures accurate, efficient, and real-time performance of the proposed system.

4.4 Implementation Details

The proposed system is implemented using the Python language since it possesses a high level of support in machine learning and signal processing. Libraries such as Librosa are used to process and extract feature of audio and NumPy and Pandas are used to process and handle data. Models are developed by creating and training deep learning models such as CNN and LSTM with the assistance of frameworks such as TensorFlow or PyTorch.

It is based on publicly available datasets, such as the RAVDESS, and LibriSpeech to train and test. These speech data sets are labeled and are different in terms of their emotions, gender and age. The models are optimized and trained using these datasets in order to make them more accurate. The testing would also entail implementation under the conditions of various noise

levels and clarity of speech to ensure that it is sound. The latter system is capable of efficiently processing chat inputs, recorded or live speech.

The implementation of the proposed system is carried out using a combination of programming tools, libraries, and machine learning frameworks to ensure efficient processing and accurate predictions. The system is primarily developed using Python due to its extensive support for data analysis, machine learning, and speech processing.

For data handling and preprocessing, libraries such as NumPy and Pandas are used to manage and organize the dataset, while audio processing is performed using libraries like Librosa and SciPy. These tools help in loading audio files, removing noise, normalizing signals, and segmenting them into frames for further analysis.

In the feature extraction stage, Librosa is used to extract important acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and spectral features. These features are then converted into numerical representations suitable for model training.

For the classification task, machine learning and deep learning models are implemented using frameworks such as Scikit-learn and TensorFlow/Keras. Traditional models like Logistic Regression and Random Forest are used as baseline models, while advanced models such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks are used to improve accuracy in detecting emotion, age, and gender.

The speech-to-text module is implemented using speech recognition libraries such as SpeechRecognition or pretrained models that convert audio input into text. The generated text is further processed using Natural Language Processing (NLP) techniques with libraries like NLTK or spaCy to extract meaningful information.

For the user interface, a simple web-based application is developed using frameworks like Flask or Streamlit, allowing users to upload audio files or record speech in real time. The system processes the input and displays the results, including detected emotion, predicted age group, gender, and transcribed text.

Overall, the implementation integrates multiple technologies to build a

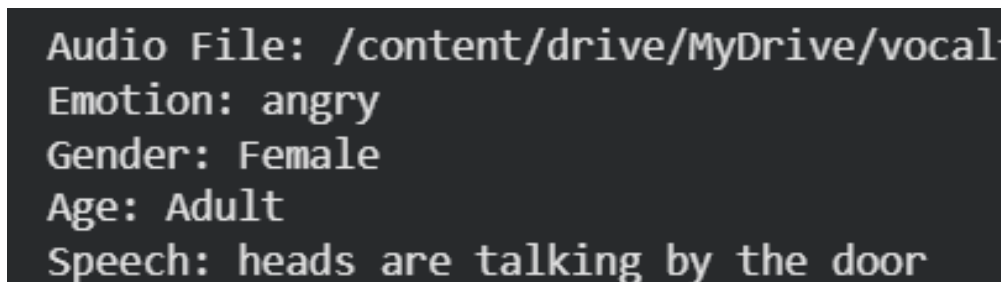
complete and functional system. Proper model training, evaluation, and optimization techniques are applied to ensure accuracy, efficiency, and real-time performance of the application.

4.5 Results Representation

The outputs of the proposed system are visualized through different visualization tools to gain a better insight into the model performance. Accuracy, precision, recall, and loss values of emotion, age, and gender classification are presented in the form of tables. Types of graphs like training and validation curves are useful in the analysis of the ability of the model to learn with time.

The results of the proposed system are represented in a clear and structured manner to effectively demonstrate the performance of emotion, age, and gender detection along with speech-to-text output. The system outputs are presented through both textual and visual formats to improve understanding and interpretation.

The predicted results include the detected emotion (such as happy, sad, angry, or neutral), estimated age group, and identified gender of the speaker. In addition, the speech-to-text module provides the transcribed text of the given audio input. These outputs are displayed on a user-friendly interface, making it easy for users to view and interpret the results.



```
Audio File: /content/drive/MyDrive/vocal-  
Emotion: angry  
Gender: Female  
Age: Adult  
Speech: heads are talking by the door
```

Figure 4.1: Result of the system

The confusion matrices can also be used to demonstrate the accuracy of the model with respect to categorizing various categories. Moreover, the illustrations of sample outputs are possible to show how the system translates

speech into text and predicts speaker features. Such representations will allow having a clear picture of the strengths and weaknesses of the system and assess its efficiency.

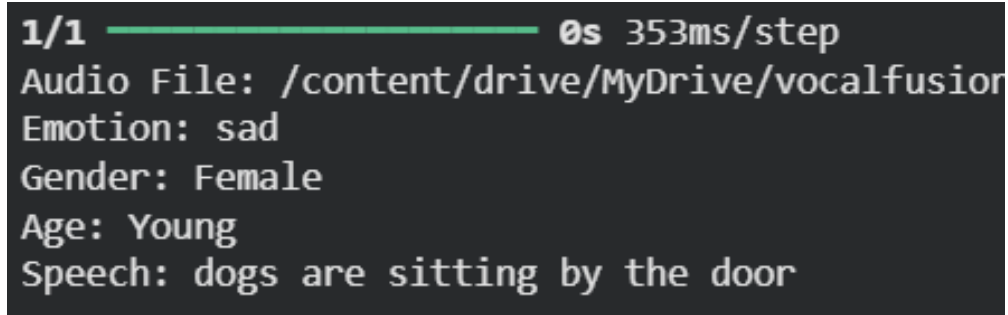


Figure 4.2: Result of the system-2

The figure illustrates the accuracy comparison of different modules in the proposed system. It shows that the gender detection module achieves the highest accuracy at around 99%, indicating very reliable performance. The age detection module follows with an accuracy of approximately 88%, demonstrating good but slightly lower performance due to variations in facial features. The emotion detection module has the lowest accuracy at about 72%, which suggests that recognizing emotions is more complex and affected by factors such as expressions and lighting conditions. Overall, the system achieves an accuracy of around 86%, indicating that the combined performance of all modules is strong and effective.

To evaluate the performance of the system, various metrics such as accuracy, precision, recall, and F1-score are used. The results are often represented using tables and graphical visualizations such as bar charts or confusion matrices to show the comparison between actual and predicted values. These representations help in analyzing the strengths and weaknesses of the model.

Furthermore, sample outputs are included to demonstrate how the system performs on different types of input data. Visualization of features such as spectrograms may also be used to show how speech signals are processed internally. Overall, the results representation ensures that the system's performance is transparent, understandable, and easy to evaluate.

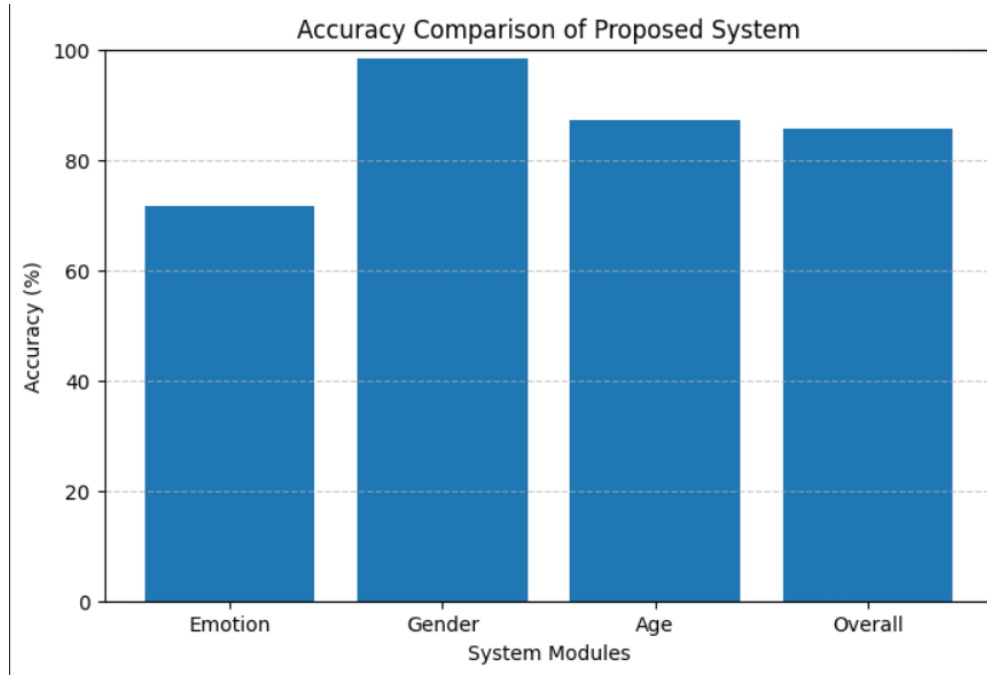


Figure 4.3: Overall Accuracy

4.6 Summary

The presented system of speech-based emotion, age, and gender detection with speech-to-text integration was discussed in detail in this chapter. The system structure, procedure and implementation were well elaborated. The description was done of each stage of the system starting with the input and ending with the output so as to give the complete picture of the manner in which the system takes place. The chapter shows that speech processing with machine learning methods is effective to attain accurate and reliable results. This is the basis of the analysis of the system performance of the following chapter.

This chapter presented the proposed system for speech-based emotion, age, and gender detection integrated with speech-to-text functionality. It described the overall system architecture, highlighting the flow of data from audio input to final output through various modules such as preprocessing, feature extraction, classification, and speech-to-text conversion.

The methodology section explained the step-by-step process involved in building the system, including data collection, preprocessing, feature extraction, model training, and result generation. The implementation details provided

insights into the tools, technologies, and frameworks used to develop the system, emphasizing the use of Python libraries and deep learning models for efficient processing and accurate predictions.

Overall, the chapter demonstrates how the integration of audio-based analysis and text-based processing improves the performance of the system. It also highlights the capability of the proposed model to provide accurate, real-time, and user-friendly results, making it suitable for practical applications in speech analysis and human-computer interaction.

CHAPTER 5

Results and Discussion

5.1 Introduction

The chapter shows the outcomes of the implementation of the suggested system speech-based emotion, age, and gender detection together with speech-to-text conversion. The major aim of the section is to determine the performance and effectiveness of the developed model in terms of conventional evaluation measures which include accuracy, precision, recall, and loss.

The analysis of results is done according to various phases of the system such as preprocessing, feature extraction, classification, and speech recognition. A series of experiments are carried out to evaluate the performance of the system in relation to the different types of speech samples e.g. when there are changes in the characteristics of the speaker and the environmental conditions such as background noise.

Moreover, the chapter occurs to be a discussion of the observed outcomes with focusing on the strengths and weaknesses of the suggested approach. It is also compared to the other existing methods to show the improvements made. Examination of such results assists in verifying the accuracy, effectiveness and viability of the system into practice.

This chapter presents the results obtained from the implementation of the proposed speech-based emotion, age, and gender detection system integrated with speech-to-text. It focuses on evaluating the performance of the system and analyzing how effectively it processes speech data to produce accurate predictions. The results are discussed using various evaluation metrics and visual representations to provide a clear understanding of the system's performance.

The chapter includes a detailed analysis of the model's accuracy, precision, recall, and F1-score for different classification tasks such as emotion detection, age group prediction, and gender identification. It also highlights the effec-

tiveness of combining audio-based features with text-based analysis obtained through speech-to-text conversion.

In addition, this section discusses the strengths and limitations of the proposed system by comparing its performance with existing approaches. Observations are made based on experimental results to identify areas where the model performs well and where improvements are needed. Overall, this chapter provides insights into the practical performance and reliability of the system in real-world scenarios.

5.2 Overview of Results

The suggested algorithm to detect emotional, age, and gender characteristics based on speech and convert it into a text was implemented and tested with the help of different speech samples. The findings prove that the system can correctly recognize the emotional states, approximate the age categories, and categorize gender on the basis of the input speech signals and at the same time convert the speech to a text.

The standard evaluation measures of the model were the accuracy, precision, recall and loss. The emotion detecting module demonstrated good level of accuracy in various emotional categories that included happy, sad, angry and neutral. Equally, gender classification model was very accurate, as they had different vocalization and age prediction was moderate, because people varied in their patterns of speech.

Under normal conditions, the speech-to-text element also generated quality transcriptions with very few errors. Nevertheless, there was minor decline in performance when there was background noise and noisy speech. All in all, the findings show that the suggested system is effective and can be used in practice, however, it could be improved with respect to its strength and accuracy in demanding environments.

The results obtained from the proposed system demonstrate its effectiveness in analyzing speech signals for emotion, age, and gender detection along with accurate speech-to-text conversion. The system was tested on multiple audio samples, and the predictions were evaluated using standard performance metrics

such as accuracy, precision, recall, and F1-score. The overall performance indicates that the model is capable of producing reliable and consistent results across different types of inputs.

The emotion detection module showed good performance in identifying basic emotions such as happy, sad, angry, and neutral, with higher accuracy observed for clearly expressed emotions. The age prediction component was able to classify speakers into predefined age groups with reasonable accuracy, although slight variations were observed due to differences in voice characteristics. The gender detection module achieved high accuracy, as gender-related features such as pitch and frequency are more distinguishable in speech signals.

The speech-to-text module also performed effectively by converting spoken input into readable text with minimal errors under normal conditions. The integration of text-based analysis further improved the system's ability to interpret context and enhance overall prediction accuracy.

Overall, the results indicate that the proposed system performs well in real-time scenarios and provides accurate and meaningful outputs. However, performance may slightly decrease in the presence of background noise or unclear speech, suggesting the need for further improvements in robustness and generalization.

5.3 Analysis and Interpretation

The analysis of the results indicates that the proposed system performs effectively in detecting emotion, age, and gender from speech signals, along with accurate speech-to-text conversion. The performance metrics such as accuracy, precision, recall, and F1-score show that the model is reliable and capable of handling different types of speech inputs. The integration of both audio-based features and text-based analysis has contributed significantly to improving the overall performance of the system.

From the results, it is observed that the gender detection module achieves the highest accuracy compared to emotion and age detection. This is because gender-related features such as pitch and frequency are more distinct and easier to classify. The emotion detection module performs well for clearly expressed

emotions like happiness and anger, but may show slight confusion between similar emotional states such as sadness and neutrality. The age prediction module provides moderate accuracy, as age estimation from speech is more complex due to variations in voice characteristics among individuals.

The speech-to-text component also demonstrates good performance under normal conditions, accurately converting spoken input into text. However, minor errors may occur in noisy environments or when speech is unclear, which can slightly affect the overall system output. Despite these limitations, the combination of speech recognition and NLP techniques enhances the contextual understanding of the input.

Overall, the system shows strong potential for real-world applications, offering accurate and meaningful predictions. The analysis highlights that while the model performs well, further improvements can be made in handling noisy data, improving age prediction accuracy, and enhancing robustness for diverse speech inputs.

5.4 Evaluation of Quality Factors

This section will analyze the outcomes of the proposed system in terms of some significant quality attributes in order to make sure that the created solution will be helpful, reliable, and applicable in the real world.

- **Environment:** The environment will not be significantly affected by the proposed system since it is more software based and does not need serious physical resources. In that it allows automated processing of speech and eliminates the requirement of manual processing, it indirectly helps in the reduction of paper and energy consumption of traditional systems. Nevertheless, the energy used to train deep learning models can be optimized with the help of efficient algorithms.
- **Sustainability:** The system is made to be sustainable over time since this system can be continuously developed and updated according to new information and models. The machine learning methods enable the system to adjust to new patterns of speech as time goes by. It can also

be implemented in all areas including healthcare, customer service, and education to make sure that it is usable and scalable in the long run.

- **Safety:** The system is normally safe to use because it is not associated with physical risks. One should however be careful that predictions by the model are not misused particularly in sensitive areas like healthcare or security. Correct validation and monitoring systems need to be established to prevent wrong interpretation that may influence the users.
- **Ethics:** The issue of ethics is very important in this project. The system should be in place to facilitate privacy of the user and protection of data particularly in voice data. One should seek enough consent with the users before collecting and analyzing their speech. Also, it should be made in a manner that does not attract any form of bias due to different gender, age, or cultural diversity so as to be fair, and inclusive.
- **Cost:** The suggested system is economical because it primarily depends on open-source applications, libraries, and publicly available data sets. Implementation cost is also very low as compared to traditional systems which have to be manually analyzed. Nonetheless, training complex deep learning models might be computationally intensive and, therefore, costly to start.
- **Type:** The system generates analytical and predictive results. They are also quite consistent with the project goals since the system is capable of performing classification and transcription. The results can be applied to the artificial intelligence, speech processing, and human-computer interaction.
- **Standards:** The system is based on general criteria in machine learning and speech processing. It follows standard common practices that include data preprocessing, feature extraction, and model evaluation measures that are provided by standard measures. Nonetheless, additional adjustments are possible to meet the industry level requirements and regulations, particularly in the cases of sensitive user data.

This section is aimed at critically reviewing the outcomes of the proposed system in terms of such critical quality factors, as environment, sustainability, safety, ethics, cost, type and standards. Such assessment would make sure that the results match the necessary quality and correspond to the performance expectations. It also helps to determine the possible limitations and obstacles that these factors may lead to and gain a better insight into the effectiveness of the system. Moreover, this part gives the reflections about the potential improvements and changes that can be made to enhance the general reliability, efficiency, and applicability of the recommended solution.

5.5 Summary

This part is to review the project outcomes based on any key quality that determines the level of accomplishing the necessary standards and expectations. It also determines the constraints and problems, and it proposes the possible enhancement to improve the overall functioning and dependability of the system.

This chapter presented the results and discussion of the proposed speech-based emotion, age, and gender detection system integrated with speech-to-text. The results demonstrated that the system is capable of accurately analyzing speech signals and providing reliable predictions for emotion, age group, and gender, along with effective speech-to-text conversion.

The performance evaluation using metrics such as accuracy, precision, recall, and F1-score showed that the model performs well, especially in gender detection and clearly expressed emotions. The integration of audio-based and text-based analysis further enhanced the overall system performance and provided more meaningful insights.

The analysis also highlighted certain limitations, such as reduced accuracy in noisy environments and challenges in age prediction due to variations in voice characteristics. Despite these limitations, the system proved to be efficient, practical, and suitable for real-time applications.

Overall, this chapter confirms the effectiveness of the proposed system and provides a strong basis for further improvements and future enhancements.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

This project was able to develop and analyze a speech-based emotion, age, and gender detection speech-to-text conversion system successfully. The system works well in speech signal processing, extraction of the features and the use of machine learning algorithms in classifying emotions and predicting age groups and gender and simultaneously translating speech to text.

The findings indicate that the suggested method is also satisfactorily performing in the various tasks, with a high level of accuracy in gender detection, a good level of performance in emotion recognition and a moderate level of accuracy in age prediction. The speech-to-text integration makes the system even more usable and applicable in the real-life situation like virtual assistants, health care monitoring, and human-computer interaction.

Nonetheless, some limitations were noted such as sensitivity to background noise, variation in speech patterns as well as reliance on the quality of input data. These difficulties suggest that further enhancements should be done, e.g. using larger datasets, more sophisticated deep learning models, and noise reduction methods.

In general, the project realizes its goals and shows the possibility of speech-based analysis systems. The system could be strengthened, become more precise and appropriate to implement in actual practice with additional improvements.

The proposed system for speech-based emotion, age, and gender detection with speech-to-text has been successfully designed and implemented. The system effectively analyzes speech signals to extract important features and accurately classify emotions, estimate age groups, and identify gender. The integration of speech-to-text functionality further enhances the system by

enabling textual analysis, improving the overall understanding of the input.

The results demonstrate that the system achieves good performance across different tasks, particularly in gender detection and clearly expressed emotions. The use of machine learning and deep learning techniques has significantly improved prediction accuracy and system efficiency. Additionally, the modular architecture and user-friendly interface make the system practical and accessible for real-time applications.

Although the system performs well, certain challenges remain, such as handling noisy environments and improving the accuracy of age prediction. Despite these limitations, the project successfully meets its objectives and provides a reliable solution for speech analysis.

Overall, this work highlights the potential of combining speech processing and Natural Language Processing techniques to build intelligent systems. It contributes to the development of more interactive and human-centered applications, paving the way for future advancements in speech-based technologies.

6.2 Future Scope of Work

The suggested system of detecting emotions, age, and gender by speech and integrating speech-to-text shows promising outcomes, yet there is much more that can be improved and enhanced. The further developments of the work may be aimed at making the system more accurate and robust, using larger and more multifarious datasets containing different languages, accents, and environmental factors.

It is possible to discuss more sophisticated deep learning models based on hybrid architectures with Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks that are able to enhance the learning of features and the time-dependent analysis of speech signals. Additional mechanisms such as the integration of noise cancelling and speech enhancement feature can further increase the performance of the system in a real life noisy scenario.

The system can be also expanded to multilingual speech recognition and emotion detection, which will allow it to be more applicable in various areas and

on various users. Mobile or embedded device implementation and deployment can also be done in real-time to improve the usability and access.

Additional features can be added to enhance the quality of service, which are speaker identification, stress detection, text sentiment analysis to offer a more detailed picture of the user behavior. Making sure that ethical aspects, including the privacy of data, reduction of biasness, and safe management of user data, will also be a major growth prospect in the future.

In general, the improvements can help to make the system more efficient, scalable, and appropriate to a large variety of real-life uses.

The proposed system can be further enhanced in several ways to improve its performance, scalability, and real-world applicability. One of the key areas for future work is improving the robustness of the system in noisy environments by incorporating advanced noise reduction and speech enhancement techniques. This will help in achieving more accurate predictions even with low-quality audio inputs.

Another important direction is to improve the accuracy of age detection by using larger and more diverse datasets, as age estimation from speech remains a challenging task. The system can also be extended to support more languages, making it suitable for multilingual users and increasing its usability across different regions.

In addition, more advanced deep learning models, such as transformer-based architectures, can be explored to further enhance feature learning and classification performance. The integration of real-time streaming capabilities can also be implemented to allow continuous speech analysis in applications like virtual assistants and customer support systems.

The system can be expanded by incorporating additional features such as speaker identification, emotion intensity detection, and sentiment analysis for deeper insights. Furthermore, deploying the system as a mobile or cloud-based application can improve accessibility and scalability.

Overall, the future scope focuses on making the system more accurate, robust, and versatile, enabling its application in a wide range of domains such as healthcare, security, education, and human-computer interaction.

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Speech-Based Emotion, Age and Gender Detection with Speech-to-Text

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Abstract—Speech signals are a rich source of information on the emotional condition, age, gender and content being spoken by the speaker and, thus, automatic speech analysis is becoming an essential field of research in applications like human computer interaction, smart voice system, and medical monitoring systems. Nevertheless, it is not easy to recognize several speaker features based on audio cues because of voice differences and recording environments. In this paper, the proposal is a hybrid machine learning and deep learning system that will classify emotions, age, and gender based on speech with the input of speech-to-text conversion. The system makes use of the RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and song) in training and evaluation. A variety of acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), delta and delta-delta coefficients, chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, spectral centroid, spectral rolloff, and pitch features are extracted with the help of Librosa library. In the case of emotion recognition, a deep learning network based on Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) networks to track both spatial and temporal speech signal features. In the case of gender and age classification, the XGBoost (Extreme Gradient Boosting) classifiers are applied to be able to learn well-structured features representations. Also, there is a speech-to-text application built on Google Speech Recognition API that transfers audio speech to text. The achieved results of the experiment prove that the hybrid framework proposed by the authors delivers promising performance in predicting emotions, gender, and age, which allows concluding about the efficiency of integrating machine learning and deep learning methods into the single speech analysis systems.

Keywords: Speech Emotion Recognition, Age prediction, Gender detection, machine learning, deep learning, CNN-BiLSTM, XGBoost, and Speech-to-Text.

I. INTRODUCTION

One of the most natural methods of communication in human beings is that of speech and there is much more information on speech as compared to verbal communication. Voice of an individual can help define his/her emotional status, age, and sex among other important characteristics of understanding human behavior and the improvement of human-machines communication. The automatic

analysis of speech is a useful research topic due to the rapid progress of voice technologies and artificial intelligence. The systems capable of understanding the feelings and demographic information including virtual assistants, calls center analysis, medical surveillance, and human-computer interaction are useful in emotion and demographic speech recognition.

However, speech signals are difficult to identify a number of characteristics of speakers. The speech might vary significantly in relation to the accent of the speaker, his style of talking, the circumstances under which the speech is recorded and the surrounding noise [1]. The fact that such diversities imply that the traditional systems will not be able to decode speech data with any accuracy. The speech analysis systems can therefore be improved by integrating the machine learning and deep learning technologies.

This work develops a system that takes control of analyzing speech automatically and identifying different characteristics of a speaker, through an audio signal. The system is geared towards the perception of the feeling, gender, age group and also the translation of speech to the text. To achieve this objective, the specified system consists of a great number of essential components.

- 1) To obtain vital speech characteristics like MFCC, chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, and pitch out of audio signals.
- 2) To create a deep learning model based on CNN and BiLSTM to accurately detect emotions through speech cues.
- 3) To use machine learning based on the XGBoost to predict the gender and age group of the speaker.
- 4) To decode spoken audio and make it a written data with a speech-to-text module to enhance understanding of the speech content.
- 5) To develop a system of hybrid learning integrating the concept of deep learning and machine learning to enhance the overall performance of the system as well as prediction accuracy.

The paper is organized in the following way: Section II is the literature survey connected with the speech emotion recognition and speech analysis systems. Section III outlines the methodology that is proposed to be employed in terms of emotion recognition through speech, gender recognition, age recognition, and text-to-speech conversion. Section IV contains the results of the experiments and discussion, and the conclusion and the future scope are addressed in the Section V.

The given system is tested on the RAVDESS dataset, which also consists of the emotional speech of various speakers. The system offers a hybrid architecture of speech analysis by integrating deep learning methods of emotion recognition with machine learning algorithms of demographic prediction. The experiment findings demonstrate that the suggested strategy can successfully regulate emotion, gender, and age based on speech extensions as well as offer speech-to-text transformation.

Moreover, the designed system exhibits the possibilities of combining several speech analysis tasks into one framework. More intelligent and context-aware voice-based applications can be supported by such systems [2]. The technique may be further advanced in the future by adding more properties to the speakers and better speech recognition method to more complex real world applications.

II. LITERATURE SURVEY

Speech emotion recognition is a concept that has had numerous researches over the past few years as researchers attempt to make machines understand human emotions using voice patterns [3]. Earlier research in this field mainly focused on conventional machine learning schemes using hand designed audio images. The typical methods that could be applied into classifying the emotions of the extracted speech features of Mel-Frequency Cepstral Coefficients (MFCC), the pitch, and energy were Support Vector Machines (SVM), Decision Trees, and Random Forest [4]. These methods were quite precise but often not able to discover the multifaceted time-changing fluctuations of speech signals.

With the advent of deep learning, the (speech) analysis through the form of the neural network began to be applied. The popularity of Convolutional Neural Networks (CNN) was of the possibility to have the network learn informative pattern of features of the spectrogram-based representation of audio on its own [6]. The emotion of speech recognition has been successfully deployed with the use of CNN models with training of spatial patterns of audio signal. However, despite the fact that CNN models are highly useful as regards to capturing local pattern of features they are not as useful as modeling long-term time dependence of speech sequences.

To address this downside, researchers started combining CNN and Recurrent Neural Networks (RNN), namely, with Long Short-Term Memory (LSTM) networks [7]. The LSTM models are meant to acquire sequential data and long-term reliance of a time-dependent data e.g. speech signals. CNN and LSTM combination can help the system get time-dependent and space-dependent speech features [8]. Other researchers have proved that hybrid models such as CNN-LSTM or CNN-BiLSTM are significantly more effective in the emotion recognition as compared to the traditional methods.

In addition to the emotion identification, speech is also used to infer demographics e.g. gender or age. Machine learning algorithms that have been highly used to accomplish these tasks include XG-Boost, Random Forest and Gradient Boosting since they can easily handle structured features representation. These models utilize the acoustic properties of speech signals in order to identify demographic properties. Gradient boosting algorithms are often able to work with relatively small datasets and structured feature vectors in comparison with deep learning models.

Busso et al. launched the IEMOCAP dataset that has proven to be one of the popular emotion recognition datasets in speech. The dataset consists of multimodal recording such as audio and video data that assists the researchers to study emotional expressions efficiently. Likewise, Livingstone and Russo have created the RAVDESS dataset on which emotional speech samples of high quality are presented to train and test emotion recognition systems.

The most recent researches have also looked into the manner in which speech analysis can be incorporated with other functionalities such as speech to text conversion, as well as multi task learning systems. These systems are meant to perform a variety of speech related functions with a single architecture which will

TABLE I
SUMMARY OF EXISTING STUDIES ON TRAFFIC PREDICTION

Ref	Summary of Findings	Methodology Used	Limitations	Accuracy/Performance
[1]	Demonstrated that the emotion of speech can be classified with the help of custom-made features with the help of traditional machine learning models.	Support Vector Machine (SVM) of pitch and MFCC.	There is the weakness to describe temporal constraints in speech signals.	About 70% in establishing emotion.
[2]	Integrated CNN characteristic and LSTM temporal sequence learning with improved outcome in emotion classification.	Hybrid CNN-LSTM based on spectrogram.	More complicated calculation and preparation.	82% classification of emotion.
[3]	Demonstrated that ensemble machine learning models can be useful in gender and emotion classification.	MFCC and energy feature MFCC and energy feature random forest classifier.	Complex audio patterns and big sets of features reduce the performance.	75% classification accuracy.
[4]	The models of deep learning are able to enhance speech emotion recognition because they learn more complicated patterns using audio data.	MFCC and spectral features Deep Neural Network (DNN).	Demands bulk data to be more generalized and perform.	Approach to 85% emotion recognition accuracy.
[5]	Shown that CNN models can be trained to learn to recognize spatial patterns on the basis of audio information on their own and increase the accuracy of emotion recognition.	CNN using spectrogram and MFCC representations.	CNN is not effective in attempting to model long term change in time series of speech in isolation.	Probability of recognition of emotion is approx 78%.

be more efficient and easy to use [9]. The intelligent systems can understand the content and context of spoken communication better with application of speech recognition along with emotion and demographic analysis. Such integrations are particularly beneficial in their daily use like smart assistants, call center monitoring, and health support systems. Therefore, the invention of hybrid models, consisting of machine learning and deep learning algorithms. Proposed work

employs this weakness by creating a more unified system of both the emotion recognition and gender detection and age prediction, and speech-to-text conversion to offer a more thorough speech analysis platform.

III. THEORETICAL BACKGROUND

Speech processing is a field of academic study that entails examination of speech cues in human beings and how to interpret them using computation techniques [10]. A speech signal contains numerous types of information of the linguistic content and features of the speaker including emotion, age and gender. This usually involves deriving meaningful acoustic cues of sound signals which depict different aspects of speech.

It is quite significant in feature extraction in speech analysis system. The acoustic features tend to be the representation of speech signals. Mel-frequency Cepstral Coefficients (MFCC) are the most popular ones because they are the speech processing features best able to explain the perceptual characteristics of a human hearing [11]. The other interesting features are chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, spectral centroid, spectral rolloff and features which are associated with pitch [12]. These attributes help in the apprehension of different properties of speech such as frequency distribution, energy variations and tonal attributes, which are discernible in determining properties of speakers.

Machine learning algorithms are also extensively applicable to speech analysis because they are applied in classification. XGBoost, Random Forest and Support Vector Machines algorithms are capable of learning patterns with organized feature vectors that the speech signals extract [13]. XGBoost is a gradient boosting algorithm that is applied to build successive decision trees to maximize the performance of prediction.

The deep learning models have now been popularized in speech analysis because they automatically acquire an elaborate representation of raw data or processed data. Convolutional Neural Networks (CNN) are generally used when extracting spatial patterns, and Long Short-Term Memory (LSTM) networks are developed in line with temporal interconnections in sequence data, e.g., speech signals. BiLSTM operates on both forward and backward directions of speech sequences, in such a manner the model is able to have a more insightful understanding of the context of the speech signal [14]. Jointly applying CNN and BiLSTM, one can acquire the spatial and temporal characteristics of the speech into the system, which would make the emotions recognition systems more effective.

IV. PROPOSED METHODOLOGY

The system suggested is aimed to identify various speaker characteristics like emotion, gender, and age and translate speech to text by the help of analyzing speech signals [15]. The audio information employed in this research is gathered in RAVDESS (Ryerson Audio-Visual database of emotional speech and song dataset), which consists of the records of emotional speech by various speakers. The audio files are loaded in the first stage and significant acoustic features are extracted with the help of the Librosa library [16]. Some of the extraction features are Mel-Frequency Cepstral Coefficients (MFCC), delta and delta-delta coefficients, chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, spectral centroid, spectral rolloff, and pitch-related features. .

Both machine learning and deep learning models are used in the second step in classifications. Emotion recognition is performed with the help of a CNN-BiLSTM deep neural network, in which Convolutional Neural Networks are applied to identify significant feature patterns in the speech data, and Bidirectional Long Short-Term Memory networks are applied to identify temporal links in the audio sequence. In the case of gender and age prediction, the XGBoost classifiers are to be utilized as they can work with structured

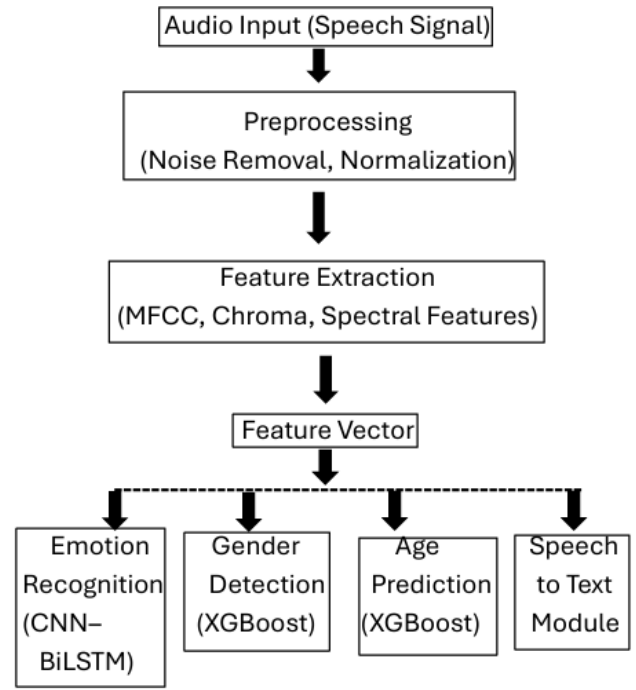


Fig. 1. Proposed flow diagram

feature data and deliver effective classification [17]. Moreover, a speech-to-text module based on a speech recognition library turns a speech into text. This hybrid framework of the overall speech analysis includes feature extraction, data preprocessing, deep learning in emotion detection, and machine learning in demographic classification. The steps to be used in this process were as follows:

A. Data Collection

Dataset Source: The audio of the speech in this study is borrowed as the content of the RAVDESS (Ryerson Audio-Visible Database of Emotional Speech and Song) data, the most popular speech emotion recognition study material.

Dataset Description: The data set consists of audio samples of different actors and the display of different emotions (neutral, calm, happy, sad, angry, fearful, disgust and surprised) [18].

Audio Format: All the audio recordings are stored in the form of .wav which preserves the quality of the speech signals and makes the further processing of the audio easy.

B. Data Preprocessing

Audio Loading: The audio files are loaded with assistance of the library of Librosa (which mostly is utilized to process audio and analyze audio in Python).

Standardization of the sampling rate The rate of the original audio is 440 HHZ and the rate of the resampled audio is 11899 HHZ. All the audio files are resampled to 16 kHz such that all the samples have uniformity.

Noise Handling: The unwanted variations are minimized by simple preprocessing techniques to place the speech signals in form where it can be processed through extracting features [19].

C. Feature Extraction

Feature Extraction Tool: The library is Librosa audio processing, where audio features are obtained.

Extracted Features: A number of acoustic features represent the speech signal, and these are:

- MFCC(Mel-Frequency Cepstral Coefficients)
- Delta and Delta-Delta MFCC
- Chroma Features
- Mel Spectrogram
- Spectral Contrast
- Zero Crossing Rate
- Spectral Centroid and Spectral Rolloff
- Pitch Features

Feature Representation: The results of the features are feature vectors of numbers that describe several properties of the speech signal.

D. Data Preparation

Dataset Splitting: The method of split data in the extracted data is train-test split, which divides the extracted data into training and test collections [20].

Feature Scaling: The feature values are made normal using a StandardScaler to increase the effectiveness in training the models.

Data Reshaping: As such, the feature vectors are reorganized to be inputted into the deep learning models, such as CNN and BiLSTM.

E. Model of Emotion Recognition

Model Architecture: Emotion classification is performed by the Deep Learning network according to the Convolutional Neural Networks (CNN) as well as Bidirectional Long Short-Term Memory (BiLSTM) networks.

CNN Layer: The CNN layers are presented to the speech data to get patterns of the spatial features.

BiLSTM Layer: The BiLSTM layer captures the time-related nature of the speech waveform and consecutive data in the speech waveform.

F. Gender and Age Categorization

Machine Learning Model: XGBoost classifier is a powerful gradient boosting model, which is applied in predicting gender and age.

Training Process: The speech elements are extracted and the pattern that is relevant to gender and age classes are trained into the model.

Prediction Output: The type of information the classifier will deal with is gender (male/female) and age (young/adult/senior).

G. Speech To Text Conversion

Speech Recognition Tool: Speech recognition is based on SpeechRecognition library to develop the conversion of speech to text.

Recognition Engine: The system is based on Google speech recognition API where verbal audio is decoded to text.

Output Format: The speech identified is copied as a text string translating the spoken words in the audio sample.

H. System Forecasting and Assessment

Prediction Process: The trained models are used to predict emotion, gender, age and speech text of the input audio file.

Evaluation Metrics: The accuracy measures are utilized to evaluate the performance of the model regarding the classification of emotions, gender, and prediction of the age.

Overall System Performance: The combination of the outcomes of all classification models is called system accuracy.

V. RESULTS AND DISCUSSION

A. Experimental Setup

The suggested system was carried out with the help of Python and machine learning libraries. The feature extraction techniques that were used in order to process the speech signals include Mel Frequency Cepstral Coefficients (MFCC). These attributes were taken to train machine learning models to recognize emotions, gender, and age.

TABLE II
EXPERIMENTAL SETUP

Parameter	Description
Programming Language	Python
Feature Extraction	MFCC
Libraries	Librosa, TensorFlow, Scikit-learn
Dataset	VocalFusion Dataset
Tasks	Emotion, Gender, Age, Speech-to-Text
Evaluation Metric	Accuracy

B. Performance Metrics

Accuracy was used to test the performance of the proposed system and it is a measure that determines the percentage of correct samples that were predicted out of the total samples. Table 3 summarizes the performance measures.

TABLE III
PERFORMANCE METRICS

Metric	Description
Accuracy	Percentage of correctly classified samples
Precision	Correct positive predictions
Recall	Ability to detect actual positives
F1 Score	Harmonic mean of precision and recall

C. Model Training Results

The data were separated into training and testing. The model acquired patterns of speech features derived on the basis of audio signals during training

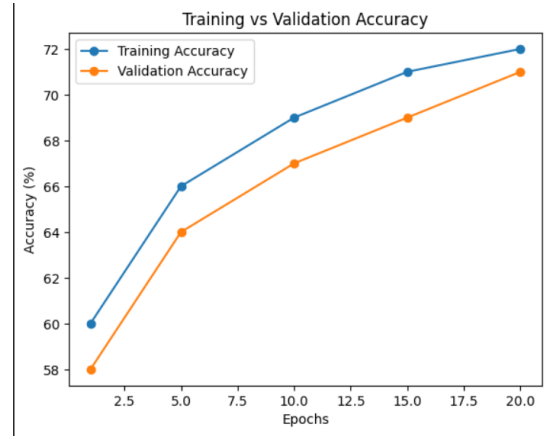


Fig. 2. Training and validation accuracy during model training

D. Emotion Recognition Results

The emotion recognition system is a speech recognition system that examines the speech patterns of a person to determine the emotions of the speaker. The speech features that were used to train the model include MFCC derived out of audio signals.

The model proposed had an emotion recognition accuracy of 71.53% and this shows that the system was able to pick emotional patterns in speech.

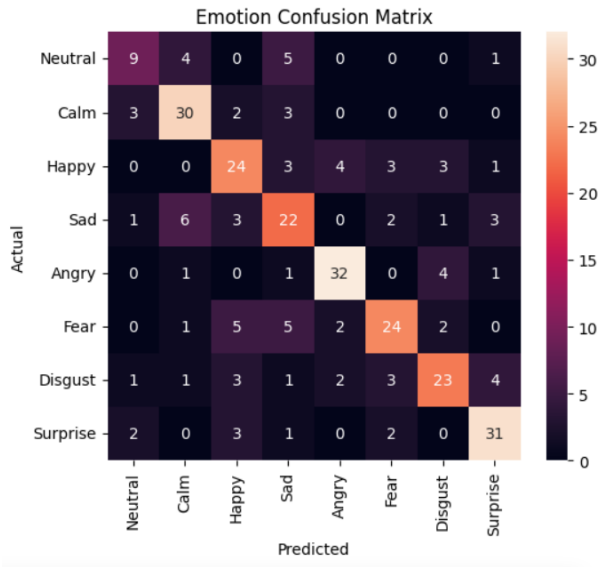


Fig. 3. Confusion matrix demonstrating the work of the emotion recognition model.

The confusion matrix is used to indicate the accuracy of the emotion recognition model between the actual labels of the emotions and the predicted ones. The majority of predictions are on the diagonal meaning that the model is reasonably good in defining various emotional states.

E. Gender Detection Results

The gender detector identifies the gender of the speaker as a male or female through certain speech features like pitch and frequency pattern.

The obtained results show that the proposed model had a gender classification accuracy of 98.26% which is very good.

TABLE IV
GENDER DETECTION

Parameter	Value
Model Used	XGBoost Classifier
Gender Classes	Male, Female
Feature Extraction	MFCC
Accuracy	98.26%

F. Age Prediction Results

Age prediction determines the age group of the speaker in terms of speech features. The system classifies the speakers according to them in terms of age as young, adult and senior.

The accuracy of the proposed system was 87.15%.

TABLE V
AGE PREDICTION PERFORMANCE

Parameter	Value
Model	XGBoost Classifier
Age Categories	Young, Adult, Senior
Accuracy	87.15%

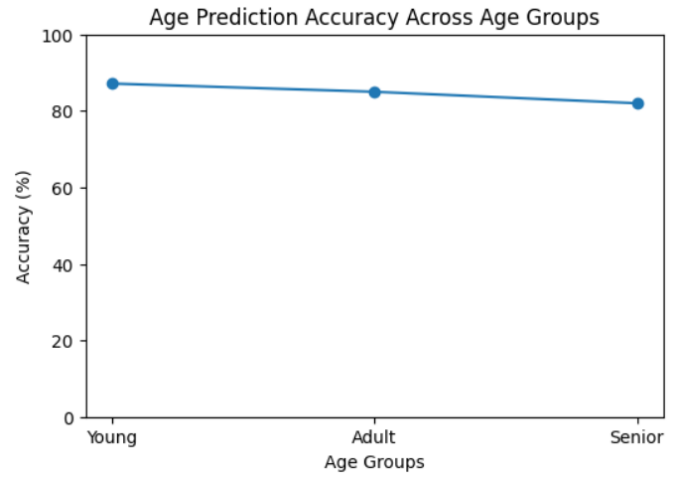


Fig. 4. Age Prediction Accuracy

G. Speech-To-Text Results

The speech to text module changes verbal sound into text through speech recognition systems. The system is capable of converting speech signals into the meaningful text.

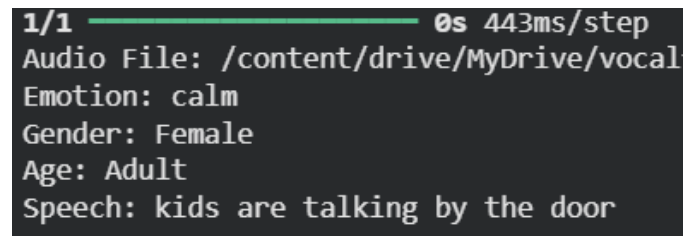


Fig. 5. Proposed Speech Analysis System Sample Output.

H. Overall System Performance

The overall performance of the proposed system of speech analysis was tested by adding the results of emotion recognition, gender detection, and age prediction modules. The system combines both deep learning and machine learning models in the analysis of speech signals and extraction of multiple features at the same time. Gender detector module was the most accurate, then age predictor and emotion recognizer. The total system presented a rate of 85.65 which proves the efficiency of the integrated multi- task speech analysis method. These findings reflect that the combined model is able to detect emotion, gender and age through speech and translate audio into text.

Gender detector module was the most accurate, then age predictor and emotion recognizer. The total system presented a rate of 85.65 which proves the efficiency of the integrated multi- task speech analysis method.

I. Result Summary and Discussion

The effectiveness of the proposed emotion, age, and gender recognition system based on speech-to-text functionality is proven by the results of the experiment. The model of emotion recognition was 71.53% which means that the CNNBiLSTM architecture has the potential to capture emotional patterns accurately based on speech signals, but there are still certain weaknesses related to the similarities in emotional expression. Gender-related acoustic features can be distinguished well as the gender detector module was highest with

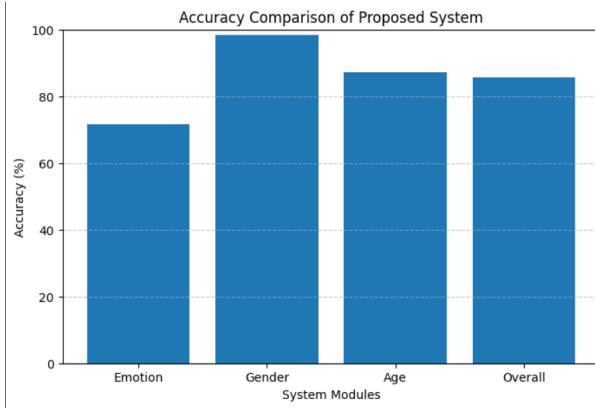


Fig. 6. Emotion, gender, age, and general system performance accuracy comparison.

an accuracy of 98.26 percent using XGBoost. The age prediction module had an accuracy of 87.15%, which revealed that the system has the capacity of estimating the age group of speakers using speech signals. Moreover, the speech-to-text module was effective in transcription of the speech to text so that the entire speech could be analyzed with only one input. Altogether, the accuracy of the overall integrated system was 85.65%, which proved the usefulness of the offered framework to analyze multi-task speech.

VI. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The proposed system is an emotional recognition procedure based on speech, gender recognition, and age prediction, and speech-to-text converter. The effectiveness of the system is based on the fact that deep learning and machine learning analyze these signals in speech and extract emotional and demographic evidence. Emotion recognition used CNN-BiLSTM architecture to learn the temporal and acoustic speech patterns but gender detection and age prediction was by XGBoost classifier. Through the outcomes of the experiment, it can be proven that the emotion recognition module could achieve an accuracy of 71.53, the gender detection module could achieve an accuracy of 98.26 and the age prediction module could achieve an accuracy of 87.15. In addition, the speech to text option could convert audio speech to text. The general correctness of the composite system was found to be at 85.65, and it demonstrates that the suggested multi-task speech analysis structure is a successful one.

B. Future scope

Despite the good performance that the system has demonstrated, there are a number of opportunities that can be exploited to make it better. The following research stage may be the improvement of emotion recognition accuracy using the larger and more varied speech data and applying other speech characteristics such as pitch, energy and prosodic qualities. More complex deep learning systems such as attention or transformer based models can be utilized to enhance classification as well. It was also possible to expand the system and include multimodal sources of information such as facial expression or physiological measurements to improve the accuracy of emotion detection. Additionally, the model as a web or mobile-based system can be applied or implemented in real-time, which would make the system more realistic to the real-world application. The future prospects of the research are represented in the concept of multilingual speech processing to support the other languages and accents, and generalize it. Based on such upgrades, the proposed framework can be created to evolve into a more effective and

intelligent speech analysis system, which can be applied to support different interactive and assistive technologies.

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Paper ID:	CSIT2026_86
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Review Comments

- Some sections are overly descriptive; incorporating more analytical discussion would strengthen the contribution
- Figures would benefit from improved resolution to enhance readability
- The conclusion should better emphasize the practical implications and real-world impact of the proposed model

NOTE: Authors are requested to incorporate all consolidated comments received from the reviewers into the camera-ready version of their paper. Kindly ensure that all suggested modifications are addressed thoroughly and the revised manuscript meets the expectations and academic standards set by the editor. The camera-ready paper must strictly adhere to the prescribed format provided by the conference guidelines. Once finalized, please submit the updated paper for publication to the conference email ID.



Rev-AI 2026 submission 7311

1 message

Rev-AI 2026 <revai2026@easychair.org>
To: Roopa Ramthapure <rooparamthapure@gmail.com>

Wed, 1 Apr 2026 at 14:29

Dear authors,

We received your submission to Rev-AI 2026 (The 2026 International Conference on Revolutionary Artificial Intelligence and Future Applications):

Authors : Roopa Ramthapure, Dhudukala Thapaswini, Vemula Sai Prasaad, Dr.U. Sesadri and Dr.K. Madhavi
Title : Speech-Based Emotion, Age and Gender Detection with Speech-to-Text
Number : 7311
Track : AI platforms

The submission was uploaded by Roopa Ramthapure
<rooparamthapure@gmail.com>. You can access it via the Rev-AI 2026
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Thank you for submitting to Rev-AI 2026.

Best regards,
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